

SCIKIT-LEARN (SKLEARN) MASTER REFERENCE GUIDE

Exhaustive Deep-Dive: Scikit-Learn Modules

Project: Universal Customer Churn & Retention Prediction System

1. Overview of Scikit-Learn

Scikit-Learn is Python's industry-standard Machine Learning library built on top of NumPy, SciPy, and Matplotlib. All algorithms follow a standard API design: `fit()` to learn patterns, and `predict()` or `transform()` to apply them.

2. Module 1: `sklearn.preprocessing.StandardScaler`

Math Formula: $z = \frac{x - \mu}{\sigma}$

- `fit()`: Computes Mean (μ) and Std Dev (σ).
- `transform()`: Scales numbers using computed (μ, σ) .
- `fit_transform()`: Performs both fitting and transformation in one step.

3. Module 2: `sklearn.model_selection.train_test_split`

Splits data into **80% Training Set** (Study) and **20% Testing Set** (Exam Vault).

Positional Unpacking Order: `X_train, X_test, y_train, y_test` (Sequence MUST be maintained).

4. Module 3: `sklearn.linear_model.LogisticRegression`

Math Formula (Sigmoid Activation): $P(y=1|X) = \frac{1}{1 + e^{-z}}$

- `model.fit(X_train, y_train)`: Adjusts internal weights (knobs) via Gradient Descent to minimize error.
- `model.predict(X_test)`: Predicts binary class outcomes (0 or 1) based on a 0.50 threshold.

5. Module 4: `sklearn.metrics` (Evaluation Suite)

- `accuracy_score()`: Proportion of total correct predictions.
- `confusion_matrix()`: 2x2 matrix of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- `classification_report()`: Detailed breakdown of Precision, Recall, and F1-Score.

6. Complete Python Pipeline Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Preprocessing Pipeline
df = pd.read_csv('data/customer_churn_data.csv')
df['Total_Charges_INR'] = df['Total_Charges_INR'].fillna(df['Total_Charges_INR'].median())
df_encoded = pd.get_dummies(df, columns=['Gender', 'Contract_Type', 'Payment_Method'])

num_cols = ['Age', 'Tenure_Months', 'Monthly_Charges_INR', 'Total_Charges_INR']
df_encoded[num_cols] = scaler.fit_transform(df_encoded[num_cols])

X = df_encoded.drop(columns=['CustomerID', 'Churn'])
y = df_encoded['Churn']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LogisticRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```